

1.

codet - Yahoo India Search Res

Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d5e/6773211f4ab787eca9d15d0

CODETANTRA Home Learn Anywhere

202501120064@mitaoe.ac.in Support Logout

1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses. The grade is determined based on the aggregate percentage.

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

- If the student passes all courses:
 - Print the aggregate percentage (formatted to two decimal places).
 - Print the grade based on the aggregate percentage.
- If the student fails any course (marks < 40 in any course), print:
 - "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

passorFa...

1 $n = \text{int}(\text{input}())$

2 $\text{marks} = \text{list}(\text{map}(\text{int}, \text{input}().\text{split}()))$

3

4 $\text{if any}(\text{mark} < 40 \text{ for mark in marks}):$

5 $\text{print}(\text{"Fail"})$

6 else:

7 $\text{total} = \text{sum}(\text{marks})$

8 $\text{max_marks} = n * 100$

9 $\text{per} = (\text{total} / \text{max_marks}) * 100$

10

11 $\text{print}(\text{"Aggregate Percentage: {per:.2f}"})$

12

13 $\text{if per} > 75:$

14 $\text{print}(\text{"Grade: Distinction"})$

15 $\text{elif per} >= 60:$

16 $\text{print}(\text{"Grade: First Division"})$

17 $\text{elif per} >= 50:$

18 $\text{print}(\text{"Grade: Second Division"})$

19 else:

20 $\text{print}(\text{"Grade: Third Division"})$

codet - Yahoo India Search Res

Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d5e/6682597fa298d347a73c1780

CODETANTRA Home Learn Anywhere

202501120064@mitaoe.ac.in Support Logout

1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

1 $\text{def fibonacci}(n):$

2

3 $\text{if } n == 1:$

4 $\text{return } 0$

5 $\text{elif } n == 2:$

6 $\text{return } 1$

7 else:

8 $\text{return fibonacci}(n-1) + \text{fibonacci}(n-2)$

9

10

11 $n = \text{int}(\text{input}())$

12 $\text{for i in range}(1, n + 1):$

13 $\text{print}(\text{fibonacci}(i), \text{end}=" ")$

14

2.

The image displays two screenshots of the CodeTANTRA web application, which is used for learning and practicing programming concepts. The interface includes a browser window at the top, a course navigation bar, and a main content area divided into a problem description and a code editor.

2.2.1. Linear search Technique

Write a program to check whether the given element is present or not in the array of elements using linear search.

Input format:

- The first line of input contains the array of integers which are separated by space
- The last line of input contains the key element to be searched

Output format:

- If the element is found, print the index. If the element found multiple times, print the starting index
- If the element is not found, print **Not found**

Sample Test Case:

Input:
1 2 3 4 3 5 6
3
Output:
2

Sample Test Cases

```
1 def linear_search(arr, key):
2     for i in range(len(arr)):
3         if arr[i] == key:
4             return i
5     return "Not found"
6 arr = list(map(int, input().split()))
7 key = int(input())
8 result = linear_search(arr, key)
9 print(result)
```

2.2.2. Captain of the Team

You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Input Format:

The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

Output Format

The output should be the height (in centimeters) of the tallest player.

Sample Test Cases

```
1 heights = list(map(int, input().split()))
2 tallest = max(heights)
3 print(tallest)
```

3.

codet - Yahoo India Search Re... Course

mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d65/698c54860aba96214c8b7d67/6774e3fff4ab787eca9da092

Ask Gemini

202501120064@mitaoe.ac.in Support Logout

3.2.1. Numpy: Matrix Operations

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:
You are required to compute and display the results of the following matrix operations:
1. **Addition** (`matrix_a + matrix_b`)
2. **Subtraction** (`matrix_a - matrix_b`)
3. **Element-wise Multiplication** (`matrix_a * matrix_b`)
4. **Matrix Multiplication** (`matrix_a · matrix_b`)
5. **Transpose of Matrix A**

Input Format:
• The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
• Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:
The program should display the results of the operations in the following order:
1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Sample Test Cases

Setting up environment...

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

• **Horizontal Stacking:** Stack the two matrices horizontally (side by side).
• **Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:
• The program should first prompt the user to input two 3×3 arrays.
• Each array consists of 3 rows, and each row contains 3 space-separated integers.
• The user will input the two arrays row by row.

Output Format:
• The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
• The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

```
1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Array2:")
8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10 # Perform horizontal stacking (hstack)
11 print("Horizontal Stack:")
12 print(np.hstack((arr1, arr2)))
13
14
15 # Perform vertical stacking (vstack)
16 print("Vertical Stack:")
17 print(np.vstack((arr1, arr2)))
18
```

codet - Yahoo India Search Res... Course

mitaoe.codetantra.com/secure/course.jsp?eucld=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d67/6774ee9df4ab787eca9da869

Ask Gemini

CODETANTRA Home Learn Anywhere 202501120064@mitaoe.ac.in Support Logout

3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user.

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using `numpy` based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

customs...

```
1 import numpy as np
2
3 # Take user input for the start, stop, and step of the sequence
4 start = int(input())
5 stop = int(input())
6 step = int(input())
7
8 # Generate the sequence using np.arange()
9 seq = np.arange(start, stop, step)
10
11 # Print the generated sequence
12 print(seq)
```

Terminal Test cases

codet - Yahoo India Search Res... Course

mitaoe.codetantra.com/secure/course.jsp?eucld=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d67/6774ef2c4ab787eca9da8c0

Ask Gemini

CODETANTRA Home Learn Anywhere 202501120064@mitaoe.ac.in Support Logout

3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

You are given two arrays `A` and `B`. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

- Arithmetic Operations:**
 - Compute the element-wise sum, difference, and product of the two arrays.
- Statistical Operations:**
 - Calculate the mean, median, and standard deviation of array `A`.
- Bitwise Operations:**
 - Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: `A1` OR `B1`).

Input Format:

- The first line contains space-separated integers representing the elements of array `A`.
- The second line contains space-separated integers representing the elements of array `B`.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

```
1 import numpy as np
2
3 def array_operations(A, B):
4
5     # Convert A and B to NumPy arrays
6     A = np.array(A)
7     B = np.array(B)
8
9     # Arithmetic Operations
10    sum_result = A + B
11    diff_result = A - B
12    prod_result = A * B
13
14    # Statistical Operations
15    mean_A = np.mean(A)
16    median_A = np.median(A)
17    std_dev_A = np.std(A)
18
19    # Bitwise Operations
20    and_result = A & B
21    or_result = A | B
22    xor_result = A ^ B
23
24    # Output results with one space between each element
25    print("Element-wise Sum:", ' '.join(map(str, sum_result)))
26    print("Element-wise Difference:", ' '.join(map(str, diff_result)))
27    print("Element-wise Product:", ' '.join(map(str, prod_result)))
28
29    print(f"Mean of A: {mean_A}")
30    print(f"Median of A: {median_A}")
31    print(f"Standard Deviation of A: {std_dev_A}")
32
33    print("Bitwise AND:", ' '.join(map(str, and_result)))
```

Terminal Test cases

4.

4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9
10 # Find the month with the highest total sales
11 df["Date"] = pd.to_datetime(df["Date"])
12 df["Month"] = df["Date"].dt.to_period("M")
13 df["Total Sales"] = df["Quantity"] * df["Price"]
14 monthly_sales = df.groupby("Month")["Total Sales"].sum()
15 # Find the month with the highest total sales
16 best_month = monthly_sales.idxmax()
17 highest_sales = monthly_sales.max()
18
19 print(f"Best month: {best_month}")
20 print(f"Total sales: ${highest_sales:2f}")
21
```

codet - Yahoo India Search Res...Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?eucld=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d6a/67861553a50fe56b8dcf9a6e

CODETANTRAHomeLearn Anywhere

202501120064@mitaoe.ac.inSupportLogout

4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	28	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	12	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 # write the code..
10 city_sales = df.groupby('City')['Quantity'].sum().reset_index()
11 best_city_row = city_sales.loc[city_sales['Quantity'].idxmax()]
12 best_city = best_city_row['City']
13
14 # Display the result
15 print(f"City sold the most products: {best_city}")
16
```

codet - Yahoo India Search Res...Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?eucld=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d6a/6788eea89ff79b6ea9e3ece1

CODETANTRAHomeLearn Anywhere

202501120064@mitaoe.ac.inSupportLogout

4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pairs that were sold most frequently.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	28	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	12	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Explanation:

Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B
- 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

- Product A and Product B: Appear in transactions on 2025-01-01 and 2025-01-03.
- Product A and Product C: Appear in transactions on 2025-01-02 and 2025-01-05.
- Product B and Product C: Appear in transactions on 2025-01-04.

Most Frequent Product Combinations:

- Product A and Product B (2 times)
- Product A and Product C (2 times)

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

frequent...

```
1 import pandas as pd
2 from itertools import combinations
3 from collections import Counter
4
5 # Prompt user to input the file name
6 file_name = input()
7
8 # Read data from the specified CSV file
9 df = pd.read_csv(file_name)
10
11 # write the code
12 grouped = df.groupby("Date")["Product"].apply(list)
13
14 pairs_counter = Counter()
15 for products in grouped:
16     pairs = combinations(sorted(products), 2)
17     pairs_counter.update(pairs)
18
19 max_frequency = max(pairs_counter.values())
20 most_common_pairs = [(pair, freq) for pair, freq in pairs_counter.items() if freq == max_frequency]
21
22 # Output the most frequent product pairs
23 for (product1, product2), frequency in most_common_pairs:
24     print(f"({product1} and {product2}): {frequency} times")
25
26 # Output the most frequent product pairs
```

Codetrantra - Yahoo India Search
+ + +

[C+DETANTRA](#) Home Learn Anywhere

202501120064@mitaoe.ac.in Support Logout

5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

- Use **30 bins** for the histogram.
- Set the **edge color** of the bars to **black** (k).
- Label the x-axis as '**Age**' and the y-axis as '**Frequency**'.
- Add the title "**Age Distribution**" to the histogram.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,1,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,235,5
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.283,C85,C
3,1,1,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,S
4,1,1,"Votralle, Mrs. Jacques Heath (Lily May Peel)",female,39,1,0,113803,53.1,C123,S
5,0,1,"Allen, Mr. William Henry",male,35,0,0,374646,8.45,S
6,0,3,"Moran, Mr. James",male,0,0,338677,0.4583,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Malson, Hester, Gusta Leonard",male,2,1,1,949999,21.075,S
9,1,1,"Johnson, Oscar M (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.4708,,C
    
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

Explorer

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15 plt.hist(data['Age'], bins=30, edgecolor='k')
16
17
18
19 plt.title('Age Distribution')
20
21
22
23
24 plt.xlabel('Age')
25
26
27
28 plt.ylabel('Frequency')
29
30
31 plt.show()
32 # Write your code here for Histogram
    
```

Debugger

Terminal Test Cases

< Prev
Reset
Submit
Next >

codet - Yahoo India Search Res

Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?euid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d6d/67e28d6e7c0b912d860e75...

mitaoe.codetantra.com

Home Learn Anywhere

202501120064@mitaoe.ac.in Support Logout

5.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:
1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to 'bar'.
3. Add the title 'Survival Count' to the chart.
4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,5
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O 1181282,7.925,,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,5
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,5
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,53.8025,446,5
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,5
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,5
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15 survival_counts = data['Survived'].value_counts()
16 survival_counts.plot(kind='bar')
17
18 plt.title('Survival Count')
19
20 plt.xlabel('Survived')
21
22 plt.ylabel('Count')
23
24 plt.show()
25 # Write your code here for Bar Plot for Survival Rate
26
27
```

Terminal Test cases

codet - Yahoo India Search Res

Course

Ask Gemini

mitaoe.codetantra.com/secure/course.jsp?euid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d6d/67e28fb27c0b912d860e7db4

mitaoe.codetantra.com

Home Learn Anywhere

202501120064@mitaoe.ac.in Support Logout

5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:
1. Group the data by the 'Sex' column, then use the 'value_counts()' function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a 'stacked bar chart' to display the survival counts.
3. Add the title 'Survival by Gender' to the chart.
4. Label the x-axis as 'Gender' and the y-axis as 'Count'.
5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,5
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O 1181282,7.925,,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,5
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,5
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,53.8025,446,5
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,5
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,5
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Gender
17 survival_counts = data.groupby('Sex')['Survived'].value_counts().unstack()
18 survival_counts.plot(kind='bar', stacked=True)
19
20 plt.title('Survival by Gender')
21 plt.xlabel('Gender')
22 plt.ylabel('Count')
23 plt.legend(['Not Survived', 'Survived'])
24 plt.show()
25
26
27
28
```

Terminal Test cases